Whether or not you trust early universe theories, taking vacuum decay seriously depends on placing a great deal of trust in the Standard Model of particle physics, which we know cannot be the whole story. Dark matter, dark energy, and the incompatibility of quantum mechanics and general relativity all point to there being something more to the universe than what we can currently write down. Whatever comes along to replace the Standard Model might, by the by, save us from even having to vaguely worry about a wayward bubble of quantum death.

Or it may be that extensions of fundamental physics present entirely new ways for the universe to end. The possibility of extra dimensions of space—the same ones that tantalize collider physicists hoping to make miniature black holes—extends the universe into new realms of the unknown. Like any explorer reaching the edges of the map, we reach out not knowing what we might find. Higher dimensions of space might allow us to solve some long-standing problems with our theories of gravity, but they also come with a warning, scrawled in the margins of the ever-growing cosmic map: here be monsters.